

Revealing Regional Economic Sentiment Beyond National Aggregates: A News-Based Measure Using a Fine-Tuned Financial Language Model

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ABSTRACT

This paper develops a high-frequency regional news sentiment index for Incheon—a metropolitan city in Korea with a distinct industrial structure—using economic news articles and a fine-tuned financial language model, KR-FinBERT. To address the limitations of conventional regional economic indicators—such as publication lags and limited timeliness—we construct a sentence-level sentiment classification framework based on 20,000 manually labeled news sentences. The BERT-based classifier substantially outperforms lexicon-based and traditional machine learning models in sentiment classification tasks. Using the predicted sentiment, we construct daily and monthly Incheon News Sentiment Index (INSI) measures and examine their empirical relevance for regional economic condition. The results show that the INSI is positively correlated with both national news sentiment indices and conventional economic sentiment indicators, exhibiting a lead of several months over regional consumer and manufacturing sentiment. Furthermore, impulse response analysis indicates that positive shocks to the INSI are associated with persistent increases in regional economic activity and prices. By focusing on Incheon as a case study, we demonstrate that region-specific sentiment indices can capture localized economic dynamics that are not reflected in national aggregates, highlighting their potential as real-time monitoring tools for regional economic conditions.

KEYWORDS

Regional News Sentiment Index, Transformer-Based NLP, Financial Language Model, Regional Economic Monitoring

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1. INTRODUCTION

Recent advances in natural language processing (NLP) have transformed unstructured text, particularly news articles, into a key source of economic information that can capture real-time sentiment more effectively than conventional indicators. While conventional measures and survey-based sentiment indices remain important, they are limited by publication lags, high costs, and difficulties in reflecting rapidly changing expectations (Ludvigson, 2004). In contrast, text data provide a rich proxy for agents' expectations and uncertainty (Gentzkow et al., 2019). The literature has evolved from dictionary-based approaches (Loughran and McDonald, 2011) and rule-based methods such as VADER (Hutto and Gilbert, 2014) to transformer-based models that better capture contextual meaning (Mishev et al., 2020). Empirical evidence shows that news-based measures improve macroeconomic forecasting and are especially valuable during periods of economic disruption (Bybee et al., 2020; Kalamara et al., 2022; Ashwin et al., 2024; Shapiro et al., 2022), with recent work further enhancing their usefulness through machine learning, deep learning, and text-augmented factor models (Seo et al., 2024; Seo, 2025; Sun et al., 2025). Overall, this literature highlights the growing importance of news-based textual data and advanced NLP methods as timely complements to traditional economic indicators.

Regional economic conditions could be heterogeneous, yet most existing text-based sentiment measures rely on aggregate national indicators, which obscure local dynamics and create information frictions. At the same time, regional hard data—such as GRDP—are limited by publication lags and weak representativeness, leading to significant information gaps (Ludvigson, 2004; Seo et al., 2022). Although prior studies highlight the importance of regional heterogeneity (Quigley, 1998) and its predictive value (Filippou et al., 2024), this dimension remains insufficiently explored in measurement frameworks. This limitation is particularly critical given that hard data are relatively scarce at the regional level, implying a greater role for soft data such as news-based sentiment. Nevertheless, existing research largely focuses on national aggregates, failing to develop region-specific news-based sentiment measures that reflect local heterogeneity.

To address these gaps, this study aims to construct a regional news-based economic sentiment index using economic news articles from Incheon, a major metropolitan city in Korea,

chosen for its distinct regional economic characteristics.¹ By leveraging high-frequency text data, the proposed index complements the limitations of conventional regional indicators and provides a real-time tool for local economic assessment and policy decision-making.

To this end, we collect economic news articles published between 2020 and 2024 from national, economic, and regional daily newspapers. The extraction of Incheon-related news articles is based on the presence of the term ‘Incheon,’ the names of administrative districts within Incheon, and the names of companies headquartered in Incheon. To infer the economic sentiment of these news articles, we employ pretrained NLP models.² Considering the economic context, we fine-tune KR-FinBERT (Kim and Shin, 2022), a financial language model trained on large-scale Korean financial texts, such as economic news articles and corporate disclosures. We demonstrate the fine-tuned BERT³ sentiment classifier outperforms dictionary-based approaches as well as traditional machine learning and deep learning methods. Based on this sentiment classifier, we construct an Incheon News Sentiment Index (INSI) and demonstrate that the INSI is related to economic sentiment indicators and real economic indicators of the Incheon region.

The main contributions of this study are as follows. First, we construct a regional economic sentiment classifier through a novel application of KR-FinBERT—a transformer-based language model pretrained on Korean financial texts—offering a more contextually sensitive approach than conventional lexicon-based and machine learning methods. By systematically adapting this domain-specific pretrained model to regional sentiment classification, the fine-tuned classifier yields substantially improved performance over these benchmarks. Second, we apply the

¹ Incheon is a key logistics and industrial hub located in the capital metropolitan area of Korea. It provides a relevant case for regional analysis due to its diverse economic structure, including aviation, port logistics, manufacturing, and bio industries. By focusing on Incheon, the study captures region-specific economic dynamics that are not fully reflected in national indicators.

² Pretrained NLP models are language models that are trained in advance on large-scale text corpora to learn general linguistic patterns and semantic representations. These models can then be fine-tuned or directly applied to downstream tasks, such as sentiment analysis, with relatively limited task-specific data.

³ Bidirectional Encoder Representations from Transformers (BERT) is a pre-trained language model proposed by Devlin et al. (2019). It is based on a Transformer encoder architecture that enables bidirectional contextual understanding. The process of further training the model to adapt it to a specific domain is referred to as fine-tuning, and in this study, the fine-tuned BERT model is applied to sentiment classification.

fine-tuned classifier to construct a regional news sentiment index for Incheon and demonstrate that it exhibits distinct dynamics from national-level sentiment indices. This finding suggests that region-specific news-based indicators capture localized economic conditions that are not reflected in national aggregates. Lastly, as an analytical tool, the INSI demonstrates substantive empirical value for understanding regional business cycles. Specifically, we show that the INSI exhibits a leading relationship with conventional regional economic indicators, enabling early detection of cyclical turning points. Moreover, by virtue of its news-based nature, the INSI provides interpretable signals that can help identify the sources of regional economic fluctuations, offering analytical advantages over traditional survey-based measures.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and methodology for constructing the regional news sentiment index. Section 4 presents the empirical results and assesses the validity of the index. Section 5 concludes the paper.

2. LITERATURE REVIEW

This study relates to three main strands of literature. The first strand addresses the use of text data as an alternative source of economic information, emphasizing its ability to overcome the limitations of traditional economic indicators. The second strand reviews recent developments and the practical usefulness of news text analysis in economic forecasting. The third strand examines the limitations of existing studies in the development of regional economic indicators, highlighting the heterogeneity of local economic structures and their responses to economic shocks. Finally, building on these strands, this study outlines its specific contributions to literature.

The first strand of the literature highlights the inherent limitations of conventional economic indicators and the merits of using text data in economic analysis. While measures such as GDP, consumer price indices, unemployment rates, and industrial production provide indispensable objective assessments of economic outcomes, they are released with substantial publication lags, involve high survey and administrative costs, and fail to capture real-time changes

in economic sentiment and expectations (Ludvigson, 2004).

As an alternative, a growing body of literature has turned to textual data—such as news articles, legislative records, corporate disclosures, and social media—as proxies for economic sentiment, expectations, and uncertainty. Gentzkow and Shapiro (2010) demonstrated that text can be transformed into quantifiable high-dimensional data, while Baker et al. (2016) established the Economic Policy Uncertainty (EPU) index by counting newspaper articles that jointly contain terms related to the economy, policy, and uncertainty, showing that text-based measures capture policy-related risks not reflected in financial variables such as the VIX. Subsequent studies by Fraiberger (2016), Thorsrud (2020), and Nyman et al. (2021) provided empirical evidence that news-based textual indicators significantly improve the forecasting performance of macroeconomic variables, including GDP, confirming that text constitutes an economic signal rather than noise.

The second strand of the literature documents the rapid evolution and use of text analysis methods in economics. As established by Gentzkow et al. (2019), text data constitute a rich repository of information that reflects the expectations and decision-making processes of economic agents. Early studies primarily relied on dictionary-based methods (Loughran and McDonald, 2011) or rule-based approaches such as VADER (Hutto and Gilbert, 2014). However, as demonstrated by Mishev et al. (2020), the methodological paradigm has shifted from these lexicon-based techniques toward transformer-based deep learning models, which deliver superior performance by capturing complex contextual nuances.

Among these, the introduction of BERT by Devlin et al. (2019) represents a major breakthrough, enabling bidirectional contextual understanding and significantly improving performance in classification tasks, particularly in sentiment analysis. However, BERT models pretrained on general corpora may fail to fully capture the specialized language of finance and economics, motivating the development of domain-specific models. Araci (2019) and Yang et al. (2020) proposed FinBERT models trained or further pretrained on financial news and corporate disclosures. They reported that FinBERT achieves an accuracy of 97% on the Financial PhraseBank dataset, substantially outperforming general-purpose BERT and traditional machine

learning methods. In the Korean context, Kim and Shin (2022) developed KR-FinBERT, a domain-specific pretrained language model trained on large-scale Korean financial news and analysts' reports, optimized to capture the nuanced semantics of Korean economic texts.

The empirical usefulness of news-based sentiment has been well documented in the economic forecasting literature. Early studies primarily rely on dictionary-based approaches to quantify sentiment from news data. Bybee et al. (2020) show that the structure of news text closely reflects key macroeconomic factors and provides additional predictive power beyond traditional numerical variables. Shapiro et al. (2022) develop a daily news sentiment index based on 24 major newspapers and show that it significantly enhances the predictive power of traditional survey-based indicators. Kalamara et al. (2022) and Ashwin et al. (2024) find that news-based indicators are particularly valuable during periods of economic distress—such as the 2008 financial crisis or the COVID-19 pandemic—when conventional models struggle to capture abrupt changes.

More recent studies, in contrast, adopt machine learning and deep learning approaches to extract sentiment from news text. In line with this trend, the Economic Statistics Department of the Bank of Korea has published a weekly news sentiment index since 2022 using a transformer-based KoNS classifier, achieving high predictive accuracy in the Korean language (Seo et al., 2024). Sun et al. (2025) improve the accuracy and stability of business cycle predictions by combining news narrative sentiment with explainable machine learning. Seo (2025) proposes a “Text-Augmented Factor Model (TFM)” and emphasizes the importance of using ubiquitous news text for early economic diagnosis, particularly when official statistics are unavailable.

The third strand of the literature addresses the limited research on measuring news-based economic sentiment at the regional level. A key concern is that national-level indices tend to obscure region-specific economic structures, as well as the heterogeneity in economic shocks and responses across regions. Quigley (1998) argues that the diversity of cities and regions constitutes a fundamental driver of growth, implying that aggregate signals may fail to capture localized vulnerabilities. Consistent with this perspective, Filippou et al. (2024) use FinBERT to analyze the Federal Reserve's Beige Book and find that regional sentiment heterogeneity provides an early signal of national recessions, even when aggregate indicators remain stable.

Recent literature further underscores the importance of localized news narratives in capturing regional economic dynamics. Ozgun and Broekel (2021) show that innovation- and technology-related news is strongly embedded within specific geographical contexts. Building on this idea, Broekel (2025) introduces the concept of the “heartbeat of regions,” demonstrating that local news sentiment can capture high-frequency fluctuations in regional economic activity more effectively than annual administrative data. However, despite growing recognition of regional heterogeneity, existing studies remain limited in assessing the practical policy applicability of such indicators, particularly in regional contexts characterized by distinct economic structures.

Several studies have attempted to construct regional news sentiment indices for Gangwon and Jeonbuk provinces in Korea (Moon et al., 2023; Lee et al., 2022). Nevertheless, research at the regional level remains at an early stage. Existing approaches largely rely on keyword matching or traditional machine learning techniques, which limit their ability to accurately capture contextual meaning and to fully leverage the sophistication of domain-specific pretrained language models such as KR-FinBERT (Kim and Shin, 2022).

This study aims to address these gaps in three key respects. First, it enhances methodological rigor beyond prior regional studies by employing a state-of-the-art KR-FinBERT model fine-tuned on a large-scale regional news dataset. Second, it implements a refined regional filtering strategy to reflect Incheon’s distinct industrial characteristics, as one of Korea’s most specialized regional economies, particularly in logistics, biotechnology, and advanced manufacturing. Finally, rather than merely constructing an index, the study empirically examines its dynamics with real economic variables—such as production, consumption, and prices—using local projections. Through this approach, it provides a robust real-time monitoring tool that can strengthen data-driven policy analysis at the local government level.

3. DATA and METHODOLOGY

3.1. Data Construction

Economic news articles convey the state of the economy and the perceptions of economic agents

in a timely manner, thereby contemporaneously influencing their decision-making processes. To capture private sector economic sentiment, we collect economic news articles from ten newspaper outlets—including comprehensive, economic, and regional newspapers.⁴ The sample covers the period from January 2020 to December 2024. Economic news articles are identified based on the “Economic” category in the BIG KINDS database.

However, national-level news sentiment fails to adequately represent the economic conditions of specific regions such as Incheon metropolitan city in Korea. Therefore, we selectively extract and analyze news articles directly related to the Incheon regional economy using conventional filtering strategies that include not only the regional name “Incheon,” but also the names of its administrative districts and major locally headquartered firms, following Moon et al. (2023).

Specifically, we identify scraped news articles related to Incheon based on the following criteria. First, we classify an article as Incheon-related if the term “Incheon” appears in the main text and no other metropolitan city names—namely Seoul, Busan, Daegu, Gwangju, Daejeon, Ulsan, or Sejong—are mentioned. Second, we classify an article as Incheon-related if it contains the term “Incheon” together with the names of its administrative districts (Jung-gu, Dong-gu, Michuhol-gu, Yeonsu-gu, Namdong-gu, Bupyeong-gu, Gyeyang-gu, Seo-gu, Ganghwa-gun, or Ongjin-gun), while excluding references to other metropolitan cities. Third, regardless of the presence of regional keywords, we consider an article as Incheon-related if it mentions a large corporation headquartered in Incheon. As of 2024, 37 firms headquartered in Incheon are included among Korea’s top 1,000 companies by domestic sales.⁵ Representative examples include firms operating in the biopharmaceutical, petrochemical, automotive manufacturing, heavy industry, and steel sectors. This approach is essential for capturing subtle regional economic signals that may otherwise be obscured by the dominant narratives of national-level media coverage.

⁴ These outlets include The Chosun Daily, Korea JoongAng Daily, The Dong-A Ilbo, Hankyoreh, The Korea Economic Daily, Maeil Business Newspaper, Incheon Ilbo, The Kyeongin Ilbo, Giho Ilbo, and The Joongboo.

⁵ This information is based on data from the Incheon Chamber of Commerce and Industry (Changes in Incheon-based firms among Korea’s top 1,000 companies ranked by domestic sales).

Table 1 reports the total number of news articles and the number of Incheon-related articles identified for the construction of the Incheon news sentiment index. Applying the three criteria described above, we extract a total of 40,927 Incheon-related news articles over the period from 2020 to 2024, corresponding to an annual average of 8,185 articles and an average share of 7.07%.

<Insert Table 1 about here>

3.2. Sentiment Classification and Labeling Procedure

Developing the INSI requires sentiment classification at the sentence level in news articles. In doing so, sentiment classifiers can be broadly classified into (i) lexicon-based and (ii) machine learning-based methods. The primary objective of the study is to fine-tune and implement the BERT model for sentiment inference, given its strong performance in classification tasks among machine-learning-based methods. We employ the lexicon-based approach solely as a benchmark for comparison.

Sentiment classification tasks based on machine learning and deep learning, such as BERT, rely on supervised learning, in which models are trained on sentiment-labeled data. In the study, we use 20,000 news sentences annotated with sentiment labels to train sentiment classification models. Following the findings of Kim et al. (2024) and Seo et al. (2024), which report that Korean Transformer-based models achieve the highest performance in sentiment classification tasks, a BERT-based model is adopted as the main model for sentiment prediction, while other algorithms are employed for validation and comparative performance evaluation. Specifically, we employ KR-FinBERT, a Korean pretrained model specialized in the economic and financial domain.⁶

News articles are first segmented into sentences, and each sentence undergoes

⁶ KR-FinBERT is a BERT-based model developed by SNUNLP (Seoul National University NLP Group) and is pre-trained on financial news articles and reports, making it well suited for economic and financial text analysis. For a detailed explanation, see Kim and Shin (2022).

preprocessing steps, including morphological analysis and vectorization, to convert the text into numerical representations. The 20,000 labeled sentences are divided into training (80%) and validation (20%) datasets. Then, we train and evaluate various algorithms—including logistic regression, random forest, support vector machines (SVM), LSTM, and BERT—to compare their performance with that of BERT. Through the fine-tuning procedure, the BERT can be adapted to a specific domain, achieving state-of-the-art performance among recent natural language processing models. We apply the trained BERT model to unlabeled news sentences to predict sentiment polarity and aggregate the resulting scores at daily, monthly, and annual levels to construct the index. Detailed technical descriptions of the fine-tuning procedure are provided in Appendix B.

Sentiment classification models typically rely on supervised learning, which requires labeled data for both training and evaluation; therefore, sentiment labels must be assigned to a subset of news text in advance. Accordingly, we select 20,000 sentences from 1,000 news articles and employ a professional annotation service to label each sentence as very positive, positive, neutral, negative, or very negative. Among the 20,000 labeled sentences, we have 13 annotators independently evaluate 4,000 sentences, while 17 annotators evaluate the remaining 16,000 sentences. We determine the final sentiment label for each sentence by majority voting and assign sentences with tied votes to a neutral label.

The label guidelines in Table 2 are based on Shapiro et al. (2022) and Seo et al. (2024), and emphasize the overall tone of the sentence rather than isolated keywords or numerical indicators appearing in the text. To account for field-specific differences in sentiment expression, we provide sentiment labeling guidelines for five domains: economy, finance, industry, real estate, and consumption and entrepreneurship.

<Insert Table 2 about here>

Table 3 presents the sentence labeling results derived from the guideline-based classification, organized by category. The overall sentiment distribution is as follows: neutral (77.0%), positive (10.5%), and negative (12.5%).⁷ The proportion of sentiment-bearing sentences (positive + negative) by sector is 33.2% in the economy category, 23.8% in finance, 22.1% in industry, 23.3% in consumption, and 23.2% in Incheon, while real estate shows the lowest share at 14.6%. We also assess inter-rater reliability using multiple metrics, with detailed results reported in Appendix A. The average majority vote agreement (MVA) is 71.9%, indicating a substantial level of consensus among annotators.

<Insert Table 3 about here>

To facilitate comparison with BERT, we construct a sentiment lexicon using the widely adopted KNU Korean Sentiment Dictionary.⁸ Since the part-of-speech (POS) tags in the KNU lexicon follow the Sejong POS tagging system developed by the National Institute of Korean Language, we employ the BAREUN morphological analyzer,⁹ which adopts the same POS tagging scheme, to ensure consistent matching between words and their corresponding sentiment scores. News articles are segmented into sentences, and each sentence is morphologically analyzed to determine its sentiment polarity.

In the study, we adapt the KNU Korean Sentiment Dictionary to align with our research

⁷ We collapse the sentiment labels into positive, neutral, and negative by merging very positive and very negative into their respective categories, as such cases are extremely rare under majority voting.

⁸ The KNU Korean Sentiment Lexicon is constructed by automatically classifying definitions from the Standard Korean Language Dictionary into positive, neutral, and negative categories using a deep learning model. It is widely used in sentiment analysis research and applications and provides sentiment scores on a five-point scale ranging from very negative (−2) to very positive (+2), enabling nuanced measurement of sentiment intensity.

⁹ The BAREUN morphological analyzer is a deep learning–based model trained on approximately 78 million news articles and a lexical database containing around 800,000 words. Its main strength lies in its ability to perform accurate morpheme segmentation and POS tagging by considering contextual and semantic information.

objectives and apply it to sentiment analysis. News articles for the year 2024 are segmented according to the morphological standards used in the KNU lexicon, and frequently appearing words are extracted. Among the selected words, we manually review and refine those identified as economic terms. As a result, 141 sentiment-bearing words related to economic conditions are identified, comprising 50 very negative, 43 negative, 36 positive, and 12 very positive terms, while all remaining words are classified as neutral.

To demonstrate the practical utility of the INSI, we apply it to the case of the city of Incheon in Korea to highlight the importance of regional industrial heterogeneity. Incheon has a distinctive industrial structure, serving as a major logistics hub with both a port and an international airport, while also hosting a high concentration of advanced manufacturing industries, including biotechnology and semiconductors. This industrial structure makes Incheon a particularly relevant setting for analyzing how sector-specific dynamics and external shocks are transmitted to regional economic activity.

4. RESULTS

4.1. Evaluation of the Sentiment Classifier

The study mainly develops a BERT-based sentiment classifier model and constructs additional models based on the lexicon-based and machine learning approaches described in the previous section for comparison. We evaluate model performance using a confusion matrix,¹⁰ which compares actual and predicted values to assess diagnostic and predictive classification accuracy. For the three-class classification task, we compute evaluation metrics using a one-vs.-rest approach for each class and obtain the final values by taking the macro-average across classes. The performance metrics are defined as follows: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$; $\text{Precision} = \frac{TP}{TP+FP}$;

¹⁰ Performance evaluations from the confusion matrix are calculated based on four components: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

$$\text{Recall}=\frac{TP}{TP+FN}; \text{Specificity}=\frac{TN}{TN+FP}; \text{F1-score}=\frac{2\times(\text{precision}\times\text{Recall})}{\text{Precision}+\text{Recall}}.$$

Table 4 reports the evaluation results of the sentiment classifiers based on 10-fold cross-validation. In particular, BERT records the highest values across all evaluation metrics, demonstrating the most consistent sentiment classification performance among the estimated models. The lexicon-based method and the random forest model exhibit relatively low evaluation scores, indicating clear limitations in sentiment classification. Logistic regression and support vector machines (SVM) produce comparatively stable results; however, their F1-scores remain around 0.61, suggesting limited classification effectiveness. The LSTM model, as a deep learning approach, achieves modest improvements over traditional machine learning models. Nevertheless, BERT substantially outperforms all other models, recording scores ranging from 0.73 to 0.86 across evaluation metrics, thereby confirming its superior classification performance.

<Insert Table 4 about here>

Based on the performance evaluation results, this study employs both BERT, which achieves the highest performance, and the lexicon-based approach, which is widely used due to its simplicity despite lower performance, to classify all economic news articles into three categories: positive, negative, and neutral. Using these classifications, we construct and compare the Incheon and national news sentiment index (NSI).¹¹

4.2. Construction of Incheon News Sentiment Index

In this section, we construct the NSI using both a relatively simple lexicon-based approach and the

¹¹ Because applying preprocessing and sentiment classification models to all news articles entails substantial computational costs, we randomly sample 50% of daily articles when constructing the national news sentiment index. In contrast, for the Incheon news sentiment index, computational constraints are relatively less binding; therefore, we utilize the full set of collected articles.

high-accuracy BERT-based sentiment classification model. The index construction method follows Seo et al. (2024), while the standardization period is adjusted to reflect the data coverage of the NSI.

First, the daily NSI is computed based on news articles published during the seven days preceding each reference date, in order to account for short-term volatility. The index is then standardized over the full sample period (2020–2024) to have a mean of 100 and a standard deviation of 10 as follows:

$$INSI_t^{(daily)} = \left(\frac{X_t - \bar{X}}{S} \right) \times 10 + 100, \text{ where } X_t = \frac{\sum_{u=1}^7 P_{t-u} - \sum_{u=1}^7 N_{t-u}}{\sum_{u=1}^7 P_{t-u} + \sum_{u=1}^7 N_{t-u}},$$

$$\bar{X} = \frac{1}{|U|} \sum_{u \in U} X_u, \text{ and } S = \sqrt{\frac{1}{|U|-1} \sum_{u \in U} (X_u - \bar{X})^2}. \quad (1)$$

P_t and N_t denote the number of positive and negative sentences, respectively. U represents the set of all dates from January 1, 2020 to December 31, 2024.

Similarly, the monthly NSI is calculated based on news articles belonging to the corresponding month and is standardized using the mean ($\bar{X}$) and standard deviation (S) of the index over the sample period as follows:

$$INSI_v^{(monthly)} = \left(\frac{X_v - \bar{X}}{S} \right) \times 10 + 100, \text{ where } X_v = \frac{\sum_{v \in U_v} P_v - \sum_{v \in U_v} N_v}{\sum_{v \in U_v} P_v + \sum_{v \in U_v} N_v},$$

$$\bar{X} = \frac{1}{|V|} \sum_{v \in V} X_v, \text{ and } S = \sqrt{\frac{1}{|V|-1} \sum_{v \in V} (X_v - \bar{X})^2}. \quad (2)$$

v refers to the collection of months spanning January 2020 through December 2024, while U_v represents the collection of calendar days that fall within a v month.

Figure 1 presents the daily and monthly INSI constructed using both lexicon-based and BERT-based sentiment classifiers, comparing the indices derived from these two approaches over the period from 2020 to 2024. Both approaches exhibit broadly similar patterns, capturing consistent directional movements throughout the sample period. The correlation coefficient between the two indices is 0.80, indicating a strong positive relationship. As shown in Figure 1,

the indices decline sharply in early 2020 in response to the COVID-19 shock, followed by a gradual recovery. Thereafter, they fluctuate around the level of 100, suggesting a relatively stable trend, before entering a downward phase in the second half of 2024.

<Insert Figure 1 about here>

Figure 2 compares the monthly Incheon NSI with the national NSI. Overall, the two indices move in a broadly similar direction, reflecting common macroeconomic conditions. However, notable gaps emerge during periods when region-specific economic issues arise in Incheon.

Positive developments driving increases in the Incheon index include: (i) announcements of improved earnings by firms located in Incheon; (ii) expanded investment in industries related to Incheon; and (iii) government policy support for key local industries and tourism. In particular, positive news related to Incheon's major industries—such as semiconductors, automobiles, bio, and steel—tends to raise the INSI.¹² In addition, as Incheon hosts Incheon International Airport, positive news associated with the recovery of travel and tourism following the easing of COVID-19 effects also contributes to upward movements in the index.

<Insert Figure 2 about here>

Conversely, negative factors contributing to declines in the index primarily include: (i) concerns over deteriorating performance in key local industries; (ii) major accidents and incidents

¹² According to the Incheon Chamber of Commerce and Industry (2025), among 37 Incheon-based firms included in the keyword set, 24 firms (64.9%) belong to the manufacturing sector, and their share rises to 86.0% in terms of total sales. Within the non-manufacturing sector, construction and wholesale/retail trade account for relatively large shares.

involving casualties; and (iii) heightened domestic and external economic uncertainty. Because region-specific news accounts for a smaller share compared with nationwide coverage, the frequency of certain issues—particularly negative ones—can temporarily surge when such events occur. As a result, the regional index exhibits higher volatility than the national index.¹³

Overall, these patterns imply that, while the INSI shares common cyclical movements with the national index, it more distinctly reflects Incheon’s unique industrial composition and locally concentrated shocks. Especially, the index responds sensitively to developments in manufacturing and trade- and logistics-intensive industries, as well as to tourism-related activity linked to the airport, capturing dimensions of economic conditions that are not fully represented in aggregate national sentiment. The observed discrepancies therefore underscore the importance of incorporating region-specific information, suggesting that the INSI provides additional, complementary insights into local economic dynamics rather than merely replicating nationwide trends.

4.3. Validity Assessment of the Incheon News Sentiment Index

This section presents several empirical assessments of the validity of the INSI, examining whether the index reliably captures region-specific economic conditions in Incheon. First, Table 5 reports the correlation analysis among the lexicon-based NSI, the BERT-based NSI developed in this study, and the Bank of Korea’s NSI. The correlation between the lexicon-based and BERT-based indices is generally high for both Incheon and the national level. Specifically, the correlation coefficients between the two sentiment classifiers are 0.80 for Incheon and 0.86 for the national index. In addition, the correlation between the Incheon and national indices is relatively strong, at 0.74 for the lexicon-based model and 0.77 for the BERT-based model. Notably, the Bank of Korea’s NSI shows higher correlations with the BERT-based indices than with the lexicon-based index. The correlation coefficients between the BOK NSI and the BERT-based Incheon and national indices

¹³ Due to an attention effect, negative local events tend to generate larger downward movements in the index compared with the upward impact generated by positive news.

are 0.74 and 0.84, respectively.¹⁴

<Insert Table 5 about here>

To compare the BERT-based news sentiment with conventional economic sentiment, we examine correlations and predictive powers as measures of how well the index captures economic sentiment in regional media coverage. Figure 3 illustrates that the Incheon and national NSIs exhibit patterns broadly similar to those of the composite consumer sentiment index (CCSI) and the composite business sentiment index (CBSI).¹⁵ However, the turning points—both troughs and peaks—of the NSIs tend to appear earlier than those of the conventional economic sentiment indicators, suggesting a leading property over the business cycle. Furthermore, both in Incheon and at the national level, the CCSI appears to be more closely related to the NSIs than the CBSI.

<Insert Figure 3 about here>

Table 6 presents the cross-correlation analysis between the NSI and various economic sentiment indicators. The results indicate that the NSI generally leads the economic sentiment indicators, with lead times ranging approximately from one to six months. For the INSI, the index tends to lead regional economic sentiment indicators, although the correlation coefficients at the

¹⁴ According to Seo et al. (2024), the BOK NSI is also constructed using a Transformer-based Korean News Sentiment (KoNS) classifier, which may explain the relatively stronger association with the KR-FinBERT-based measures.

¹⁵ The CCSI is a composite index derived from six major consumer tendency survey components, including current living conditions, and is designed to comprehensively capture consumer perceptions of overall economic conditions. The CBSI, in contrast, aggregates key business tendency survey components—five from manufacturing and four from non-manufacturing sectors—to assess firms’ overall perceptions of business conditions.

maximum lead are relatively modest, ranging from 0.34 to 0.51. Given the relatively large share of manufacturing in the Incheon economy, sentiment indicators related to manufacturing sectors exhibit stronger correlations compared with those associated with non-manufacturing.

Moreover, the INSI appears to lead economic sentiment indicators with a longer lag structure compared to the national NSI. In contrast, the national NSI shows relatively shorter lead times with respect to the CCSI and exhibits comparatively stronger correlations.¹⁶

<Insert Table 6 about here>

To assess the predictive power of the NSI, we estimate the following AR(1) model:

$$Y_t = \alpha + \beta Y_{t-1} + \gamma NSI_{t-1}^{(monthly)} + \varepsilon_t \quad (3)$$

where Y_t denotes the economic sentiment indicator, Y_{t-1} is its one-period lag, and $NSI_{t-1}^{(monthly)}$ represents the one-period lagged monthly NSI. The coefficient γ captures the additional explanatory power of the NSI beyond the persistence of the dependent variable. Therefore, we assess predictive power using the statistical significance of γ .

Table 7 reports the estimation results. For the Incheon case, the coefficient on the lagged INSI is statistically significant in the regressions for the CCSI and the manufacturing CBSI at the 5% and 1% levels, respectively, but not for the non-manufacturing CBSI. This pattern is likely related to the manufacturing-intensive industrial structure of the Incheon economy.

At the national level, similar results are observed. The lagged national NSI is statistically significant for the CCSI at the 1% level and the manufacturing CBSI at the 5% level, while it is not statistically significant for the non-manufacturing CBSI. Overall, these findings suggest that

¹⁶ These findings are consistent with the cross-correlation results between the national NSI and economic sentiment indicators reported in Seo et al. (2024).

both Incheon and national NSIs have strong explanatory power and provide useful forward-looking information, particularly for consumer and manufacturing-related sentiment indicators.

<Insert Table 7 about here>

Finally, it is important to investigate how the INSI influences real economic indicators in Incheon. To this end, we examine whether the INSI affects key macroeconomic indicators by estimating impulse response functions (IRFs) that capture the dynamic effects of shocks to the INSI on real economic activity. We employ the local projection method proposed by Jordà (2005) to derive the impulse response functions.¹⁷

Formally, for each horizon h , we estimate:

$$y_{i,t+h} = \alpha_i^h + \beta_i^h NSI_t^{(monthly)} + \sum_{\tau=1}^T \alpha_{i,\tau}^h NSI_{t-\tau}^{(monthly)} + A_i^h \sum_{\tau=0}^T Y_{t-\tau} + \varepsilon_{i,t+h} \quad (4)$$

where $y_{i,t+h}$ denotes a i -th variable of the four real economic variables included in the vector Y_t . The real economic indicators for Incheon consist of (i) the cyclical component of the coincident composite index (CCI), (ii) the manufacturing production index, (iii) the large retail store sales index, and (iv) the consumer price index (CPI). All variables are transformed into logarithms prior to estimation. Given the sample size constraints, the horizon is set to a finite period of $h = 0, \dots, 12$ and the lag length is restricted to $T = 1$, and the sequence of estimated coefficients β_i^h captures the dynamic response of the i -th variable to a shock in the INSI.

Figure 4 presents the impulse response functions to a positive shock in the INSI, along with 90% confidence intervals. Specifically, a positive shock to the INSI leads to an increase in the cyclical component of the Incheon CCI, with the effect becoming statistically significant from

¹⁷ As in Jordà (2005), the local projection approach estimates the dynamic response of a variable to a shock at different horizons through a sequence of regressions and is generally considered less sensitive to model misspecification than the conventional VAR-based approach.

$h = 3$ and persisting for more than 12 months, which indicates a relatively long-lasting impact. The manufacturing production index also exhibits a positive response from $h = 1$, although its statistical significance is less pronounced compared with that of the CCI.

The large retail store sales index, which captures regional consumption activity, shows a statistically significant positive response to a shock in the INSI; however, the effect appears to be short-lived, becoming insignificant after $h = 5$. In contrast, the CPI exhibits a more persistent and highly statistically significant response, suggesting that the impact of the INSI shock on prices may endure over a longer horizon. Overall, a positive shock to the INSI generates favorable effects on real economic activity, suggesting that the resulting impulse response patterns resemble those typically associated with productivity shocks.¹⁸

5. Conclusion

This study develops the Incheon News Sentiment Index (INSI) using economic news articles related to Incheon, a metropolitan city in Korea characterized by a distinct industrial structure, and a fine-tuned financial language model—KR-FinBERT. It then examines the empirical relevance of the index for regional economic condition. The study contributes to extending analysis from national aggregates to the regional level, where empirical evidence remains limited despite substantial regional heterogeneity.

Using 20,000 sentiment-labeled sentences, we implement lexicon-based and machine learning-based approaches and demonstrate that the BERT-based sentiment classifier significantly outperforms traditional methods in terms of accuracy, precision, recall, and F1-score. Based on sentence-level sentiment classification, we construct and standardize daily and monthly Incheon and national NSI measures over the sample period from 2020 to 2024.

To assess the validity of the INSI, we conduct several validation exercises. The results

¹⁸ These findings are consistent with the impulse response results for the national NSI in the study, as well as with the evidence reported by Shapiro et al. (2022) and Seo et al. (2024).

show that the INSI exhibits moderate correlations with conventional economic sentiment indicators, while also leading regional consumer and manufacturing sentiment by several months. Furthermore, impulse response analysis shows that positive shocks to the INSI generate persistent increases in regional economic activity and prices. The estimated impulse response patterns are consistent with those typically observed following productivity-related shocks, suggesting that the INSI captures economically meaningful information about underlying real economic conditions.

More importantly, the findings highlight the relevance of regional heterogeneity in economic measurement. Unlike national-level indices, the INSI reflects the distinctive industrial structure of Incheon, including its concentration in logistics, manufacturing, and airport-related activities, and responds sensitively to region-specific developments. This result supports the view that incorporating localized information can improve the accuracy and relevance of economic monitoring, particularly in regions with diverse industrial compositions.

From a policy perspective, the proposed index provides a practical tool for real-time regional economic monitoring and can support data-driven decision-making at the local government level. In particular, given the increasing importance of timely information in macroeconomic policy, news-based sentiment indicators may complement existing survey-based measures and enhance early detection of economic turning points.

Despite these contributions, several limitations remain. First, the construction of the INSI relies on a predefined set of keywords and article selection criteria, which may introduce selection bias. Second, although the BERT-based model improves classification performance, further improvements can be achieved by expanding the training dataset or incorporating more advanced generative AI techniques. Third, while this study focuses on a single regional case, extending the framework to multiple regions would allow for a more systematic analysis of regional heterogeneity and comparative dynamics.

Future research could address these limitations by developing sector-specific sentiment indices, refining automated news filtering and topic extraction methods, and applying the proposed framework to other regions. Such extensions would further enhance the role of news-based

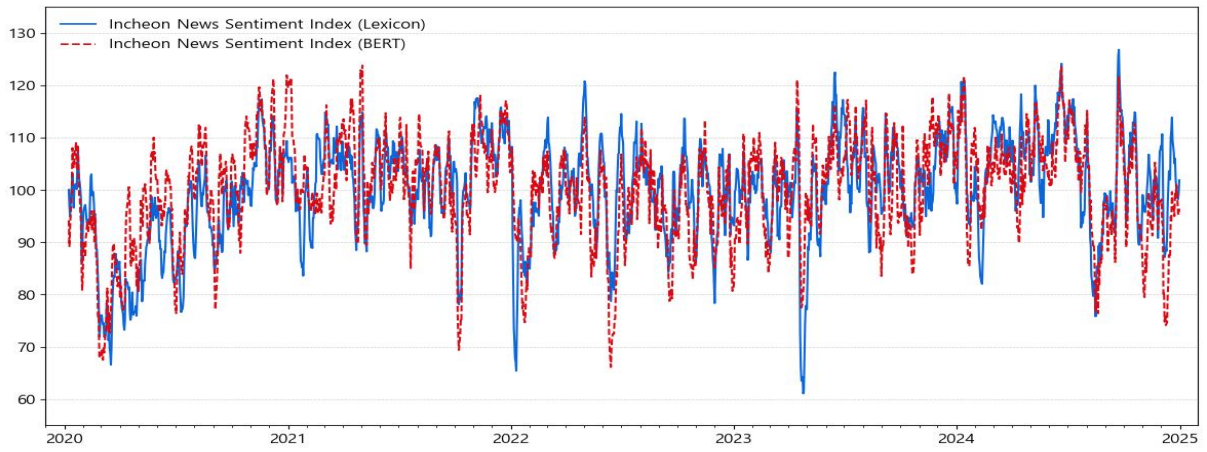
sentiment as a real-time economic monitoring tool and contribute to a deeper understanding of regional economic dynamics.

REFERENCES

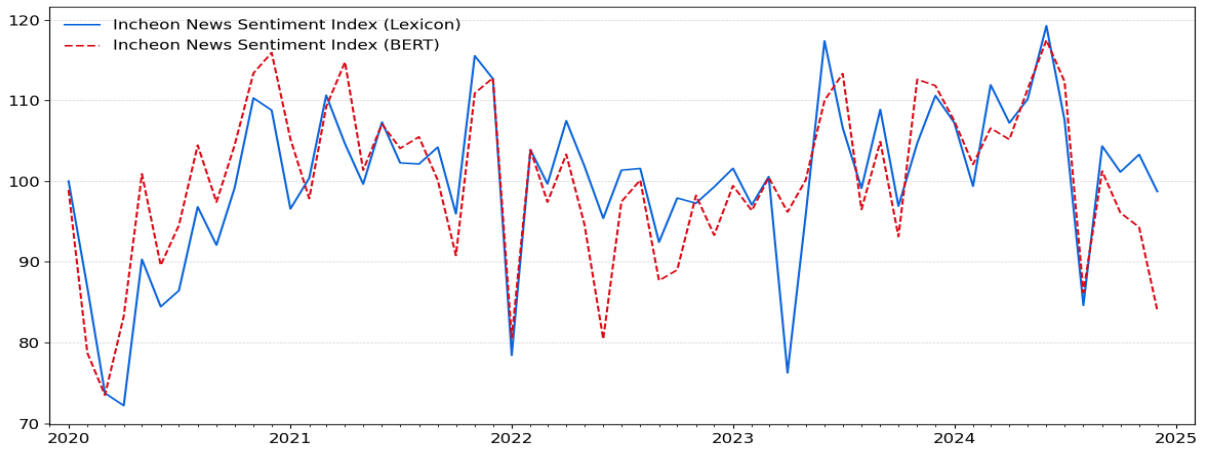
- Araci, D., 2019. FinBERT: Financial Sentiment Analysis with Pre-Trained Language Models. *arXiv preprint arXiv:1908.10063*.
- Ashwin, J., Kalamara, E. and Saiz, L., 2024. Nowcasting Euro Area GDP with News Sentiment: A Tale of Two Crises. *Journal of Applied Econometrics*, 39(5), pp.887-905.
- Baker, S.R., Bloom, N. and Davis, S.J., 2016. Measuring Economic Policy Uncertainty. *The Quarterly Journal of Economics*, 131(4), pp.1593-1636.
- Broekel, T., 2025. Feeling the Heartbeat of Regions: Local News and Economic Sentiments. *Journal of Economic Geography*, 25(6), pp.937-976.
- Bybee, L., Kelly, B.T., Manela, A. and Xiu, D., 2020. *The Structure of Economic News* (No. w26648). National Bureau of Economic Research.
- Devlin, J., Chang, M.W., Lee, K. and Toutanova, K., 2019, June. Bert: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171-4186).
- Feinstein, A.R. and Cicchetti, D.V., 1990. High Agreement but Low Kappa: I. The Problems of Two Paradoxes. *Journal of Clinical Epidemiology*, 43(6), pp.543-549.
- Filippou, I., Garciga, C., Mitchell, J. and Nguyen, M.T., 2024. Regional Economic Sentiment: Constructing Quantitative Estimates from the Beige Book and Testing Their Ability to Forecast Recessions. *Economic Commentary*, (2024-08).
- Fleiss, J.L., 1971. Measuring Nominal Scale Agreement among Many Raters. *Psychological Bulletin*, 76(5), p.378.
- Fraiberger, S., 2016, November. News sentiment and cross-country fluctuations. In *Proceedings of the First Workshop on NLP and Computational Social Science* (pp. 125-131).
- Gentzkow, M. and Shapiro, J.M., 2010. What Drives Media Slant? Evidence from US Daily Newspapers. *Econometrica*, 78(1), pp.35-71.
- Gentzkow, M., Kelly, B. and Taddy, M., 2019. Text as Data. *Journal of Economic Literature*, 57(3), pp.535-574.
- Hutto, C. and Gilbert, E., 2014, May. Vader: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. In *Proceedings of the international AAAI conference on web and social media* (Vol. 8, No. 1, pp. 216-225).
- Jordà, Ò., 2005. Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review*, 95(1), pp.161-182.
- Kalamara, E., Turrell, A., Redl, C., Kapetanios, G. and Kapadia, S., 2022. Making Text Count: Economic Forecasting Using Newspaper Text. *Journal of Applied Econometrics*, 37(5), pp.896-919.
- Kim, E. and Shin, H., 2022, "KR-FinBERT: KR-BERT-Medium Adapted with Financial Domain Data", GitHub repository, <https://huggingface.co/snunlp/KR-FinBert>.
- Kim, S., Choi, Y., Jeon, J.H. and Lu, Y., 2024. Sentiment Matters in Stock Market: Construction of Sentiment Index Using Machine Learning. *Journal of Economic Theory and Econometrics*, 35(4), pp.87-112.
- Krippendorff, K., 2011. Computing Krippendorff's Alpha-Reliability. Departmental Papers (ASC). Annenberg School for Communication, University of Pennsylvania.
- Landis, J.R. and Koch, G.G., 1977. The Measurement of Observer Agreement for Categorical Data. *Biometrics*, pp.159-174.
- Lee, K., Choi, S. and Kim, M., 2022. A Study on Developing Jeonbuk Economic Indicators by Utilizing News Big Data (뉴스 빅데이터를 활용한 전북지역경기지수 개발 연구). Jeonbuk State

- Institute. [In Korean]
- Loughran, T. and McDonald, B., 2011. When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks. *The Journal of Finance*, 66(1), pp.35-65.
- Ludvigson, S.C., 2004. Consumer Confidence and Consumer Spending. *Journal of Economic Perspectives*, 18(2), pp.29-50.
- Manning, C. and Schutze, H., 1999. Foundations of Statistical Natural Language Processing. MIT press.
- Mishev, K., Gjorgjevikj, A., Vodenska, I., Chitkushev, L.T. and Trajanov, D., 2020. Evaluation of Sentiment Analysis in Finance: From Lexicons to Transformers. *IEEE access*, 8, pp.131662-131682.
- Moon, S., Kim, S. and Lee, Y., 2023. Development and Application of the Gangwon Economic News Index (강원 경제뉴스지수의 개발 및 활용). Gangneung Branch, Bank of Korea. [In Korean]
- Nyman, R., Kapadia, S. and Tuckett, D., 2021. News and Narratives in Financial Systems: Exploiting Big Data for Systemic Risk Assessment. *Journal of Economic Dynamics and Control*, 127, p.104-119.
- Ozgun, B. and Broekel, T., 2021. The Geography of Innovation and Technology News-An Empirical Study of the German News Media. *Technological Forecasting and Social Change*, 167, p.120692.
- Quigley, J.M., 1998. Urban Diversity and Economic Growth. *Journal of Economic perspectives*, 12(2), pp.127-138.
- Seo, B., 2025. Econometric Forecasting Using Ubiquitous News Text: Text-Enhanced Factor Model. *International Journal of Forecasting*, 41(3), pp.1055-1072.
- Seo, B., Lee, Y., and Cho, H., 2022. Machine-Learning-Based News Sentiment Index (NSI) of Korea (한국의 기계학습 기반 뉴스심리지수). *Bank of Korea WP*, 15. [In Korean]
- Seo, B., Lee, Y. and Cho, H., 2024. Measuring News Sentiment of Korea using Transformer. *Korean Economic Review*, 40(1), pp.149-176.
- Shapiro, A.H., Sudhof, M. and Wilson, D.J., 2022. Measuring News Sentiment. *Journal of Econometrics*, 228(2), pp.221-243.
- Sheng, V.S., Provost, F. and Ipeirotis, P.G., 2008, August. Get Another Label? Improving Data Quality and Data Mining Using Multiple, Noisy Labelers. In *Proceedings of the 14th ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 614-622).
- Snow, R., O'connor, B., Jurafsky, D. and Ng, A.Y., 2008, October. Cheap and Fast—but Is It Good? Evaluating Non-Expert Annotations for Natural Language Tasks. In *Proceedings of the 2008 conference on empirical methods in natural language processing* (pp. 254-263).
- Sun, W., Wang, Y., Zhang, L., Chen, X.H. and Hoang, Y.H., 2025. Enhancing Economic Cycle Forecasting Based on Interpretable Machine Learning and News Narrative Sentiment. *Technological Forecasting and Social Change*, 215, p.124094.
- Thorsrud, L.A., 2020. Words Are the New Numbers: A Newsy Coincident Index of the Business Cycle. *Journal of Business & Economic Statistics*, 38(2), pp.393-409.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is All You Need. *Advances in neural information processing systems*, 30.
- Yang, Y., Uy, M.C.S. and Huang, A., 2020. FinBERT: A Pretrained Language Model for Financial Communications. *arXiv preprint arXiv:2006.08097*.

Figure 1. Incheon News Sentiment Index (INSI)



Panel (a): Daily INSI

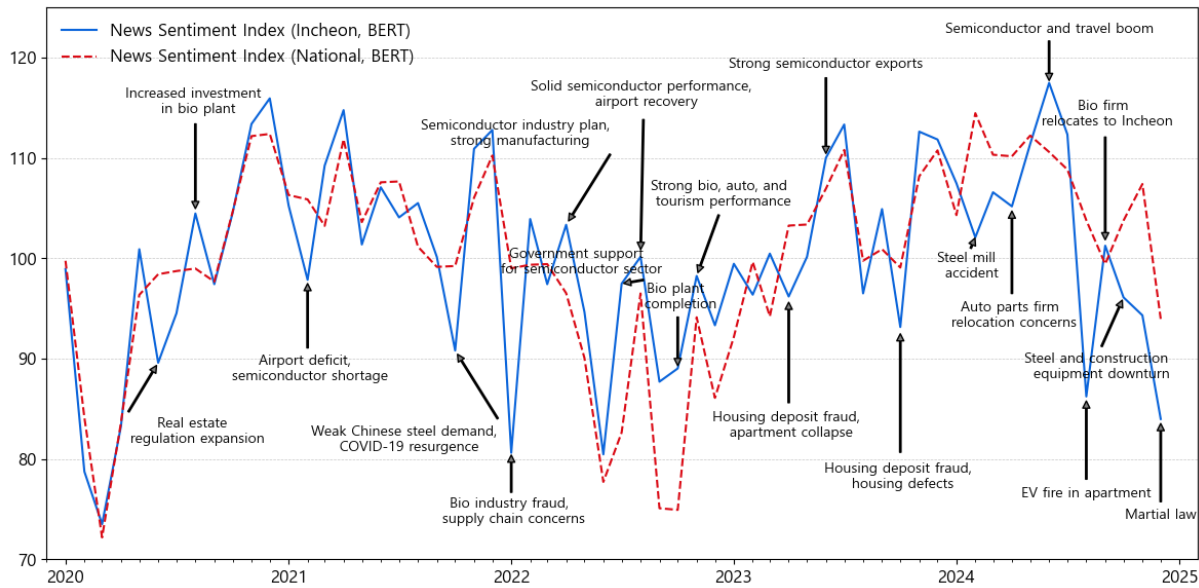


Panel (b): Monthly INSI

Note: The lexicon-based INSI is constructed using a dictionary of economic terms extracted from the KNU Korean sentiment dictionary. The BERT-based INSI is derived using a Korean version of FinBERT, i.e., KR-FinBERT (Kim and Shin, 2022), fine-tuned on economic news.

Source: Authors' calculations

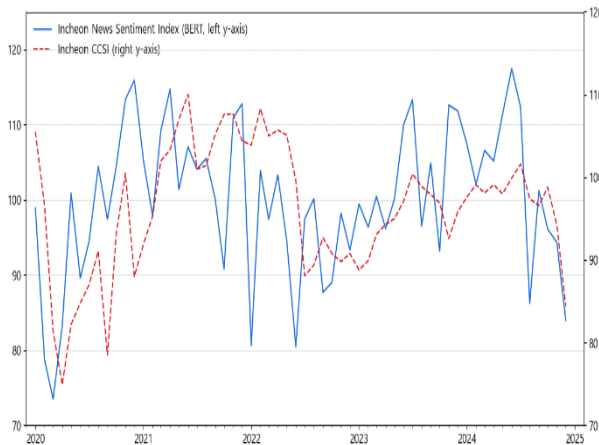
Figure 2. Incheon and National News Sentiment Indices (BERT)



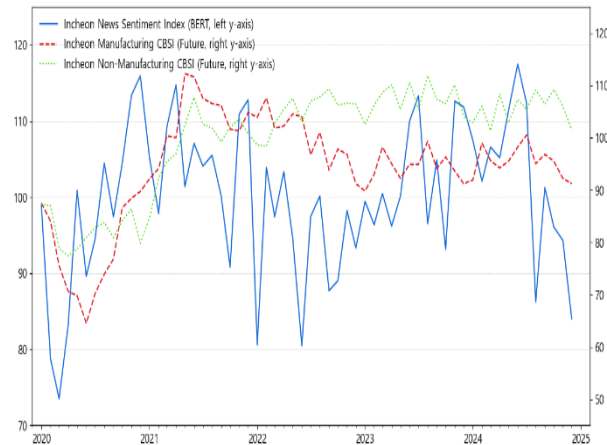
Note: We identify the factors driving the discrepancies between the Incheon and national indices by manually reviewing Incheon news articles. The BERT-based INSI is derived using a Korean version of FinBERT, i.e., KR-FinBERT (Kim and Shin, 2022), fine-tuned on economic news.

Source: Authors' calculations

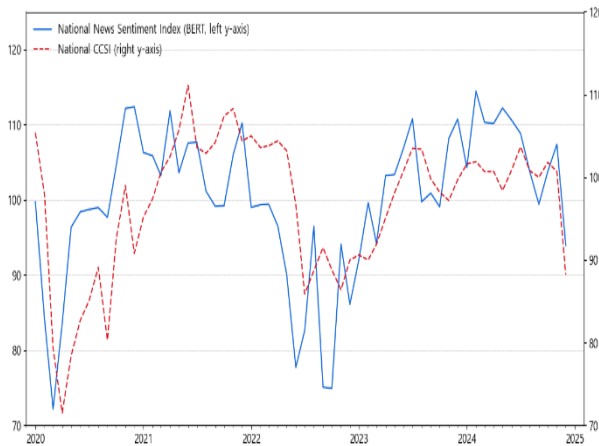
Figure 3. BERT-based News Sentiment Index (NSI) vs. Economic Sentiment Indicators



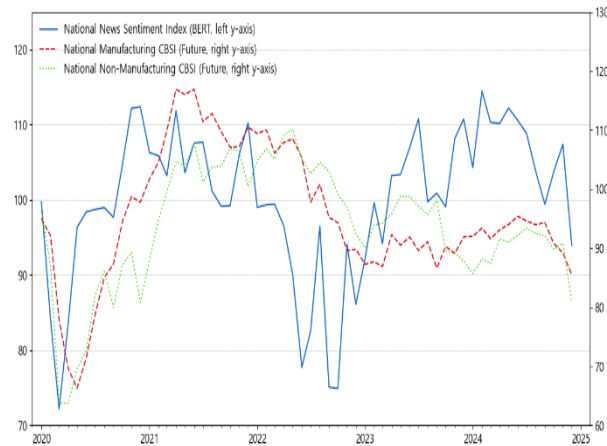
Panel (a): Incheon NSI and Incheon CCSI



Panel (b): Incheon NSI and Incheon CBSI



Panel (c): National NSI and National CCSI

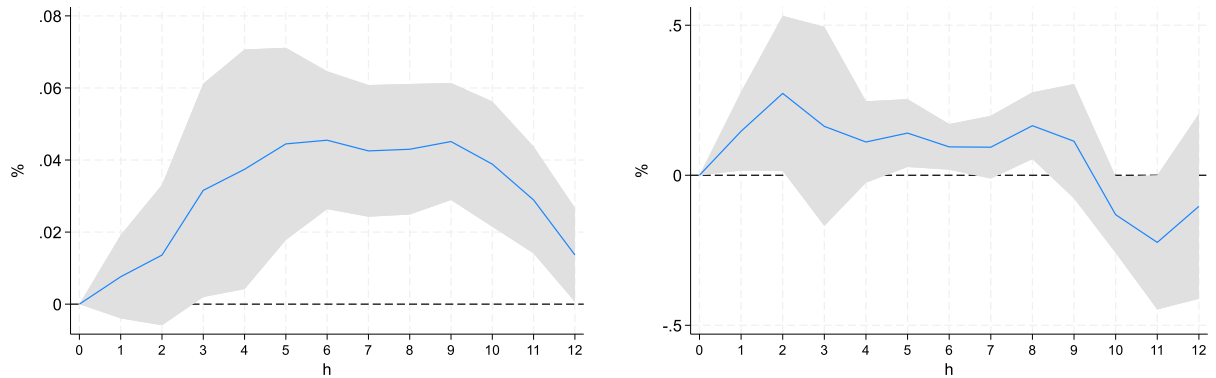


Panel (d): National NSI and National CBSI

Note: The news sentiment indices in this figure are based on BERT-based measures. CCSI and CBSI stand for the composite consumer sentiment index and the composite business sentiment index, respectively.

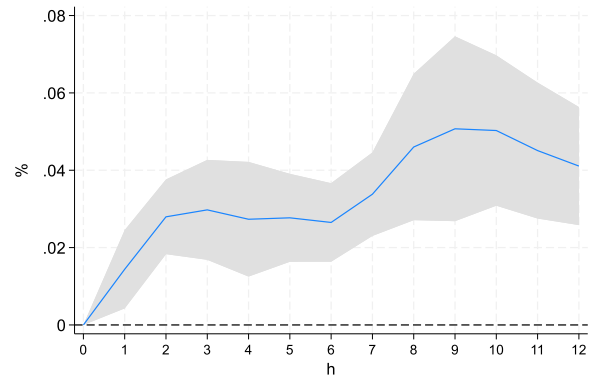
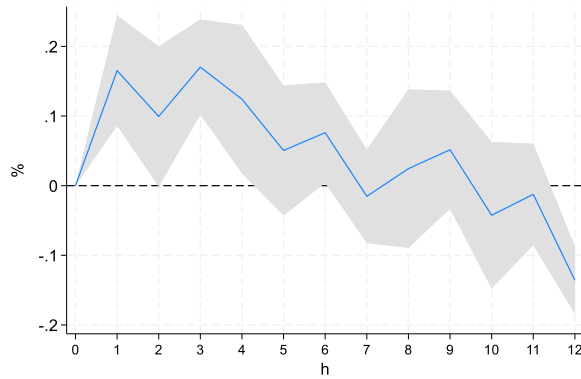
Source: Authors' calculations

Figure 4. Impulse Responses to a Shock in the BERT-based Incheon News Sentiment Index



Panel (a): Cyclical Component of the CCI (Incheon)

Panel (b): Manufacturing Production Index (Incheon)



Panel (c): Large Retail Store Sales Index (Incheon)

Panel (d): CPI (Incheon)

Note: The figures represent the responses of the economic indicators to a 1% increase (shock) in the BERT-based Incheon News Sentiment Index (INSI). The shaded areas denote the 90% confidence intervals, calculated using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors following Newey and West (1987).

Source: Authors' calculations

Table 1. News Article Collection

	2020	2021	2022	2023	2024
Total Articles	131,933	107,502	112,046	108,146	117,288
Incheon Articles	10,333	8,211	7,909	6,968	7,506
Share of Incheon Articles	7.83%	7.64%	7.06%	6.44%	6.40%

Note: Incheon-related articles are identified using keyword-based filtering: (i) articles mentioning “Incheon” without other metropolitan city names; (ii) articles co-mentioning “Incheon” and its administrative districts; and (iii) articles referring to major firms headquartered in Incheon, following Moon et al. (2023).

Source: Authors’ calculations

Table 2. Economic Sentiment Label Guidelines

General Guideline	
Question: Please select the sentiment intensity expressed in the following economic news sentence.	
Response options: (1) Very negative (2) Negative (3) Neutral (4) Positive (5) Very positive	
Annotators are instructed to focus on the journalist's tone and emotional expression rather than the underlying economic conditions described in the article. In other words, sentiment labels should reflect the author's expressive stance, not the objective content itself.	
Expressions conveying concern, doubt, or caution are interpreted as negative sentiment, whereas optimistic or confident tones are interpreted as positive sentiment.	
For example, even when an article reports objectively positive indicators such as high GDP growth, it should be labeled as negative if the text emphasizes concerns about economic overheating or includes negative sentiment expressions.	
Domain	Field-Specific Guideline
Economy	- Sentences that simply describe changes in macroeconomic indicators such as GDP or inflation are classified as neutral. - When changes in economic indicators are discussed together with specific causes or explanations, sentiment is classified as positive or negative.
Finance	- Sentences that merely report movements in stock prices, interest rates, or exchange rates are classified as neutral. - When such movements are accompanied by explicit reasons or interpretations, sentiment is classified as positive or negative.
Industry	- Advertising-related sentences are classified as neutral. - Sentences that describe past facts or events are classified as neutral. - When past events carry clear implications for the current situation or future expectations, sentiment is classified as positive or negative. - Any reference to changes in a firm's financial condition is classified as positive or negative, even if the mention is brief.
Real estate	- Apartment sales advertisements are classified as neutral. - Statements regarding unsold housing units or conditions in the construction sector are classified as positive or negative.
Consumption & Entrepreneurship	- Product advertisements are classified as neutral. - Sentences that simply describe changes in consumption- or employment-related indicators are classified as neutral. - When such changes are discussed together with specific reasons or interpretations, sentiment is classified as positive or negative.

Note: The labeling guidelines are adapted from Shapiro et al. (2022) and Seo et al. (2022) and rewritten to suit the objectives of the study.

Table 3. Sentiment Labels in Training Set

	Positive		Negative		Neutral	
	The Number of Sentences	%	The Number of Sentences	%	The Number of Sentences	%
Economy	229	10.5	496	22.7	1,457	66.8
Finance	349	9.1	567	14.8	2,927	76.2
Industry	534	10.6	580	11.5	3,923	77.9
Real Estate	168	6.1	232	8.5	2,336	85.4
Consumption & Entrepreneurship	626	14.4	390	9.0	3,341	76.7
Incheon	188	10.2	240	13.0	1,418	76.8
Total	2,093	10.5	2,505	12.5	15,402	77.0

Note: Percentages (%) represent the proportion of each domain.

Source: Authors' calculations

Table 4. Evaluation of the Sentiment Classification Model (10-Fold Cross-Validation)

	Lexicon	Logistic Regression	Random Forest	SVM	LSTM	BERT
Accuracy	0.7121 (0.0075)	0.7319 (0.0066)	0.7933 (0.0049)	0.7235 (0.0058)	0.7467 (0.0188)	0.8522 (0.0087)
Precision	0.5246 (0.0100)	0.5841 (0.0077)	0.7162 (0.0283)	0.5768 (0.0075)	0.5837 (0.0186)	0.7602 (0.0229)
Recall	0.5609 (0.0124)	0.6931 (0.0102)	0.4432 (0.0089)	0.6891 (0.0126)	0.6375 (0.0150)	0.7329 (0.0277)
Specificity	0.7738 (0.0068)	0.6822 (0.0139)	0.2251 (0.0218)	0.6882 (0.0195)	0.8093 (0.0077)	0.8581 (0.0123)
F1-score	0.5393 (0.0107)	0.6183 (0.0088)	0.4770 (0.0127)	0.6106 (0.008)	0.6045 (0.0151)	0.7435 (0.0139)

Note: Each value represents the mean and standard deviation of the performance metrics evaluated using 10-fold cross-validation. Boldface indicates the highest value for each evaluation metric. The lexicon-based INSI is constructed using a dictionary of economic terms extracted from the KNU Korean sentiment dictionary. The LSTM model adopts a bidirectional structure, allowing it to capture information from both past and future contexts (embedding dimension = 128, hidden size = 128, batch size = 32, epochs = 15, early stopping patience = 3, dropout = 0.2). The BERT-based INSI is derived using a Korean version of FinBERT, i.e., KR-FinBERT (Kim and Shin, 2022), fine-tuned on economic news. The model is trained with a learning rate of 5e-5, batch size of 32, 10 epochs, weight decay of 0.01, early stopping patience of 3, and dropout rate of 0.1.

Source: Authors' calculations

Table 5. Correlation between the Incheon/National News Sentiment Indices (NSI)

	Incheon NSI (Lexicon)	National NSI (Lexicon)	Incheon NSI (BERT)	National NSI (BERT)	National NSI (BOK)
Incheon NSI (Lexicon)	1	0.74	0.80	0.55	0.58
National NSI (Lexicon)	0.74	1	0.70	0.86	0.76
Incheon NSI (BERT)	0.80	0.70	1	0.77	0.74
National NSI (BERT)	0.55	0.86	0.77	1	0.84
National NSI (BOK)	0.58	0.76	0.74	0.84	1

Note: The lexicon-based INSI is constructed using a dictionary of economic terms extracted from the KNU Korean sentiment dictionary. The BERT-based INSI is derived using a Korean version of FinBERT, i.e., KR-FinBERT (Kim and Shin, 2022), fine-tuned on economic news. The BOK NSI is also computed using a Transformer-based Korean News Sentiment (KoNS) classifier.

Source: Authors' calculations

Table 6. Cross-Correlation between the BERT-based News Sentiment Index (NSI) and Economic Sentiment Indicators

NSI	Economic Sentiment Indicator	Lag at Maximum Correlation	Correlation coefficient
Incheon	CCSI	-6	0.51
	Living Standard of Household CSI	-6	0.43
	Domestic Economic Situation CSI	-6	0.49
	Expectations of Living Standard of Household CSI	-1	0.46
	Expectations of Domestic Economic Situation CSI	0	0.42
	Manufacturing CBSI	-6	0.51
	Manufacturing Business Condition BSI	-3	0.50
	Non-Manufacturing CBSI	-7	0.37
	Non-Manufacturing Business Condition BSI	-5	0.34
National	CCSI	-1	0.66
	Living Standard of Household CSI	-1	0.72
	Domestic Economic Situation CSI	-1	0.68
	Expectations of Living Standard of Household CSI	0	0.77
	Expectations of Domestic Economic Situation CSI	0	0.67
	Manufacturing CBSI	-3	0.45
	Manufacturing Business Condition BSI	-3	0.45
	Non-Manufacturing CBSI	-9	0.26
	Non-Manufacturing Business Condition BSI	-10	0.21

Note: Cross-correlations are computed between each BERT-based news sentiment index (NSI) and its corresponding indicators. A negative lag indicates that the NSI leads, while a positive lag indicates that the comparison indicator leads.

Source: Authors' calculations

Table 7. Estimation Results for Economic Sentiment Indicators Including the BERT-based News Sentiment Index (NSI)

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Dependent Variable: Economic Sentiment Indicator					
	Incheon CCSI	Incheon Manufacturing CBI	Incheon Non- Manufacturing CBI	National CCSI	National Manufacturing CBI	National Non- Manufacturing CBI
L,Economic Sentiment Indicator	0.690*** (0.108)	0.879*** (0.037)	0.904*** (0.044)	0.641*** (0.073)	0.908*** (0.040)	0.881*** (0.053)
L.Incheon NSI	0.176** (0.085)	0.140*** (0.049)	0.067 (0.047)			
L.National NSI				0.269*** (0.086)	0.172** (0.080)	0.091 (0.064)
Constant	11.840* (6.925)	-2.556 (6.867)	3.099 (7.534)	7.585 (6.129)	-8.581 (5.892)	1.781 (6.186)
Obs.	59	59	59	59	59	59
R-squared	0.607	0.853	0.858	0.727	0.898	0.770

Note: L. denotes the one-period lag. The estimation period spans from January 2020 to December 2024. For business sentiment, the indices in business future tendency survey are used. L. Economic Sentiment Indicator refers to the one-period lag of the dependent variable. The Incheon and national News Sentiment Indices are estimated using the BERT-based sentiment classifier. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Source: Authors' calculations

APPENDIX

A. Reliability of Sentiment Labeling

To ensure the reliability of the manually constructed sentiment-labeled dataset, we conduct a comprehensive inter-rater reliability (IRR) analysis. Given the characteristics of the dataset—comprising 20,000 sentences evaluated by 13 to 17 annotators per sentence—we validate the labeling process using a range of statistical measures.

In doing so, we use Fleiss’ Kappa (Fleiss, 1971) to measure agreement for categorical labels, which is defined as follows:²²

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \text{ where } P_i = \frac{1}{n(n-1)} \left(\sum_{j=1}^k n_{i,j}^2 - n \right), \bar{P} = \frac{1}{N} \sum_{i=1}^N P_i, \\ \mathcal{P}_j = \frac{1}{Nn} \sum_{i=1}^N n_{i,j}, \text{ and } \bar{P}_e = \frac{1}{N} \sum_{j=1}^k \mathcal{P}_j^2. \quad (\text{A1})$$

Here, N denotes the total number of sentences (20,000), n indicates the number of annotators per sentence (13 or 17), k is the number of sentiment categories (5: from very negative to very positive), and n_{ij} represents the number of raters who assigned the j -th category to the i -th sentence. Intuitively, Fleiss’ Kappa quantifies how much the observed agreement ($\bar{P}$) among annotators exceeds what would be expected by chance ($\bar{P}_e$), normalized by the maximum possible agreement beyond chance. Because the dataset contains two groups with different numbers of annotators per sentence (Group A: 13 annotators; Group B: 17 annotators), this is computed separately for each group to reflect this difference.

In addition, Krippendorff’s Alpha (Krippendorff, 2011) is a reliability coefficient developed to measure the agreement among observers, coders, or measuring instruments drawing distinctions among typically unstructured phenomena. As a generalization of several known reliability indices, α applies to any number of observers, any number of categories, and any metric or level of measurement – including nominal, ordinal, interval, and ratio scales. It is defined as:

²² Fleiss’ Kappa is one of the most established standard metrics for assessing agreement among multiple annotators on nominal scales and is widely regarded as a reliable and academically robust index.

$$\alpha = 1 - \frac{D_o}{D_e}. \quad (\text{A2})$$

where D_o is the observed disagreement among values assigned to units of analysis:

$$D_o = \frac{1}{n} \sum_c \sum_k o_{ck} \cdot \text{metric} \delta_{ck}^2. \quad (\text{A3})$$

and D_e is the disagreement one would expect when the coding of units is attributable to chance rather than to the properties of these units:

$$D_e = \frac{1}{n(n-1)} \sum_c \sum_k n_c \cdot n_k \cdot \text{metric} \delta_{c,k}^2. \quad (\text{A4})$$

Here, the arguments in the two disagreement measures, $o_{c,k}$, n_c , n_k and n , refer to the frequencies of values in coincidence matrices. The term δ_{ck}^2 represents the distance function between categories c and k , which accounts for different levels of measurement; for example, in nominal scales, $\delta_{ck}^2 = 1$ if $c \neq k$, and 0 if $c = k$.

Similar to Fleiss' Kappa, Krippendorff's Alpha measures the extent to which the observed disagreement (D_o) is reduced relative to the disagreement expected by chance (D_e). This allows for cross-validation between agreement metrics and provides a more comprehensive evaluation of reliability, particularly because α can handle incomplete or missing data. The interpretation of both indices followed the widely accepted criteria proposed by Landis and Koch (1977) in Table A8.

Table A8. Interpretation Criteria for Agreement Coefficients

Value Range	Interpretation
< 0.00	Poor
0.00 – 0.20	Slight
0.21 – 0.40	Fair
0.41 – 0.60	Moderate
0.61 – 0.80	Substantial
0.81 – 1.00	Almost Perfect

Note: These interpretation criteria are used to assess the degree of agreement for both Fleiss' Kappa and Krippendorff's Alpha coefficients.

Source: Landis and Koch (1977)

As summarized in Table A9, the Fleiss’ Kappa and Krippendorff’s Alpha scores for the dataset fall within the “slight agreement” range. However, when considering the following two statistical factors, these values suggest that a level of agreement beyond the raw coefficients is achieved under the experimental conditions of the study.

First, the observed scores can be affected by the so-called Kappa paradox (Feinstein and Cicchetti, 1990), which occurs when the distribution of categories is highly imbalanced. In our dataset, 77% of the sentences are labeled as “neutral,” resulting in a high probability of by chance agreement. This imbalance may lead to an underestimation of the Kappa coefficient.

Second, the number of annotators in the study is substantially higher than in typical studies (usually 2–3 annotators), with up to 17 annotators per sentence. According to Condorcet’s Jury Theorem, as the number of voters increases, the probability of unanimous agreement decreases, while the accuracy of the majority decision tends to converge toward the true outcome. However, this theorem assumes independence among individual judgments; in practice, such independence may not be fully satisfied when annotators evaluate the same textual data under similar conditions.

Table A9. Inter-Rater Reliability Results

Group	Metric	Score	Interpretation
Group A (13 annotators)	Fleiss’ Kappa	0.1754	Slight Agreement
Group B (17 annotators)	Fleiss’ Kappa	0.1106	Slight Agreement
Full Dataset	Krippendorff’s Alpha	0.1271	Slight Agreement

Note: Fleiss’ Kappa is computed separately for Group A and Group B due to differences in the number of annotators per sentence, while Krippendorff’s Alpha is calculated over the full dataset to account for variability in annotator counts. Interpretation follows the criteria proposed by Landis and Koch (1977).

Source: Authors’ calculations

We further employ the majority vote agreement (MVA) as a measure of practical consensus, defined as follows:

$$MVA = \frac{1}{N} \sum_{i=1}^N \left(\frac{\max(n_{i,1}, n_{i,2}, \dots, n_{i,k})}{n_i} \right), \quad (A5)$$

where $\max(n_{i,1}, n_{i,2}, \dots, n_{i,k})$ denotes the number of annotators who selected the most frequent label (i.e., the mode) for the i -th sentence, and n_i represents the total number of annotators who evaluated the i -th sentence. That is, the MVA captures the proportion of annotators who agree with the majority label for each sentence, thereby providing an intuitive measure of collective consensus.²³ Unlike chance-corrected metrics such as Fleiss’ Kappa, MVA directly reflects the extent to which a dominant label emerges within a group of annotators, regardless of category distribution.

In our analysis, the average MVA reaches 71.9%, indicating that, on average, more than two-thirds of annotators agree with the majority label for each sentence. Furthermore, over 98% of the sentences achieve a majority consensus (i.e., more than 50% agreement), suggesting that a stable collective judgment is formed in nearly all cases. Although perfect agreement across all annotators is difficult to achieve—particularly in settings involving a large number of annotators (up to 17 in the study)—these results indicate that the final labels derived from majority voting are sufficiently consistent and reliable for sentiment analysis.

²³ MVA has been widely adopted in crowdsourcing-based labeling studies since Snow et al. (2008) and Sheng et al. (2008) demonstrated the effectiveness of majority aggregation for validating data quality in non-expert annotation settings with multiple labelers.

B. BERT Model Fine-Tuning: KR-FinBERT

Pretrained language models are known to achieve the strongest performance in sentiment analysis and text classification tasks. Accordingly, we fine-tune a pretrained language model, which generally requires less training data and lower computational cost compared to other deep learning approaches. In this paper, we employ KR-FinBERT (Kim and Shin, 2022), developed by the Seoul National University NLP Group (SNUNLP), a pretrained BERT model trained on Korean economic news articles and analyst reports, and fine-tune it using the labeled training set constructed in the study.

The following provides a concise description of the BERT architecture and its fine-tuning process, adapted from Vaswani et al. (2017) and Devlin et al. (2019). Let the input consist of N sentences, where each sentence $S_j = [x_1^{(j)}, x_2^{(j)}, \dots, x_n^{(j)}]$ for $j = 1, \dots, N$ is a sequence of tokens with fixed length $n = 80$ after tokenization and padding. Each sentence is independently processed by the model, with special tokens [CLS] and [SEP] added at the beginning and end, respectively:

$$X^{(j)} = [[\text{CLS}], x_1^{(j)}, x_2^{(j)}, \dots, x_n^{(j)}, [\text{SEP}]]. \quad (\text{B1})$$

Each token $x_i^{(j)}$ for $i = 1, \dots, n$ in sentence j is represented by a combined embedding:

$$\mathbf{E}_i^{(j)} = \text{Token}(x_i^{(j)}) + \text{Segment}(s_i^{(j)}) + \text{Position}(i), \quad (\text{B2})$$

where $\mathbf{E}_i^{(j)} \in \mathbb{R}^d$ is the combined embedding of token $x_i^{(j)}$, and $d = 768$ denotes the embedding dimension. Each component is defined as follows: $\text{Token}(x_i^{(j)}) \in \mathbb{R}^d$ is the token embedding from a pretrained subword vocabulary, $\text{Segment}(s_i^{(j)}) \in \mathbb{R}^d$ encodes the segment ID with $s_i \in \{0,1\}$, and $\text{Position}(i) \in \mathbb{R}^d$ is a learned embedding for the token's position i in the sequence.

These embeddings for each sentence j are passed through L Transformer encoder layers (for $\ell = 1, \dots, L$, where $L = 12$) of the pretrained KR-FinBERT model:

$$\mathbf{H}_j^{(\ell)} = \text{Transformer}^{(\ell)}(\mathbf{H}_j^{(\ell-1)}) \text{ and } \mathbf{H}_j^{(0)} = \mathbf{E}_j, \quad (\text{B3})$$

where $\mathbf{H}_j^{(\ell)} \in \mathbb{R}^{(n+2) \times d}$ denotes the output of the ℓ -th encoder layer for sentence S_j , and $\mathbf{E}_j = [\mathbf{E}_{[\text{CLS}]}^{(j)}, \mathbf{E}_1^{(j)}, \dots, \mathbf{E}_n^{(j)}, \mathbf{E}_{[\text{SEP}]}^{(j)}] \in \mathbb{R}^{(n+2) \times d}$ is the input embedding matrix of sentence S_j .

For sentence-level classification tasks, the contextualized representation of the [CLS] token is used for each sentence.²⁴ Specifically, for sentence S_j , the output of the final (L-th) Transformer encoder layer is denoted as $\mathbf{H}_j^{(L)} \in \mathbb{R}^{(n+2) \times d}$, and the [CLS] token's representation is given by

$$\mathbf{h}_{[\text{CLS}]}^{(j)} = (\mathbf{H}_j^{(L)})_{[\text{CLS}]} \quad (\text{B4})$$

where $(\mathbf{H}_j^{(L)})_{[\text{CLS}]}$ denotes the final hidden representation of the [CLS] token for sentences S_j .

Each sentence-level [CLS] representation is then fed into a task-specific classification head. The predicted probability vector for sentence S_j is as follows:

$$\hat{\mathbf{y}}^{(j)} = \text{softmax}(W \cdot \mathbf{h}_{[\text{CLS}]}^{(j)} + b), \quad (\text{B5})$$

where $W \in \mathbb{R}^{C \times d}$ and $b \in \mathbb{R}^C$ are learnable parameters of the classification layer, and $C = 3$ is the number of target classes, with $\mathcal{C} = \{\text{positive}, \text{negative}, \text{neutral}\}$. The softmax function is defined as $\text{softmax}(\mathbf{x}) = \frac{\exp(x_k)}{\sum_{k=1}^C \exp(x_k)}$, where $\mathbf{x} \in \mathbb{R}^C$ and $\exp(\mathbf{x}) = [e^{x_1}, \dots, e^{x_C}]$. It maps a real-valued vector $\mathbf{x}$ to a probability distribution over C sentiment classes, i.e., $\hat{\mathbf{y}}^{(j)} = \text{softmax}(\mathbf{x}^{(j)}) \in [0,1]^C$ with $\hat{\mathbf{y}}^{(j)} = [\hat{y}_1^{(j)}, \dots, \hat{y}_C^{(j)}]$ and $\sum_{c=1}^C \hat{y}_c^{(j)} = 1$.

The model parameters θ are optimized via backpropagation using AdamW to minimize a weighted cross-entropy loss function that accounts for class imbalance, as follows:

²⁴ The [CLS] token is a special token that stands for "classification". The embedding corresponding to the [CLS] token is designed to capture an aggregate representation of an input sequence.

$$\mathcal{L}(\theta|\mathcal{D}) = -\frac{1}{N} \sum_{j=1}^N \sum_{c=1}^C \omega_c y_c^{(j)} \log \hat{y}_c^{(j)}, \quad (\text{B6})$$

where $\mathcal{D} = \{(S_j, \hat{y}^{(j)})\}_{j=1}^N$ with $\hat{y}^{(j)} = [\hat{y}_1^{(j)}, \dots, \hat{y}_C^{(j)}]$ denotes the training dataset, consisting of N sentence-label pairs. $\hat{y}_c^{(j)} \in \{0,1\}$ is the one-hot encoded true label for class c associated with sentence S_j and $\hat{y}_c^{(j)}$ is the predicted probability of class c for sentence S_j . The weight for class c is given by $\omega_c = \frac{N}{C \cdot n_c}$ with N being the total number of sentences, C the number of classes, and n_c the number of samples in class c . In this paper, the model is trained with a learning rate of 5e-5, a batch size of 32, 10 epochs, a weight decay of 0.01, an early stopping patience of 2, and a dropout rate of 0.1.

After fine-tuning, the trained KR-FinBERT is applied to unlabeled sentences to perform sentiment inference. For each sentence S_j , the predicted sentiment label is determined as

$$\tilde{y}^{(j)} = \arg \max_{c \in \{1, \dots, C\}} \hat{y}_c^{(j)}, \quad (\text{B7})$$

where $\hat{y}_c^{(j)}$ denotes the predicted probability of class c .